Percentile in Percentile: rating scholars against their career cohort by the impact of all their papers

Panos Ipeirotis[*†] Sen Tian[*]

Abstract

Committees often compare scholars early in their careers, before citations have accumulated, yet citation counts and the h-index drift with field, calendar and seniority, and the h-index reads only one corner of a record. We propose PiP (percentile in percentile). Each paper is ranked against papers from the same year; for every impact threshold θ, the scholar's number of papers at or above θ is ranked against peers who started in the same year; PiP averages these standings over $\theta \geq \tau$. With $\tau = 0.5$, only papers in the top half of their year can earn credit; τ is a dial between quantity and impact. We compute PiP for 11.1 million Semantic Scholar authors from the publications and citations up to each evaluation year. Paper percentiles settle quickly (correlation 0.91 between ages 3 and 10), and the distribution of PiP changes little across cohorts from 1960 to 2015, while the mean h-index at career age 10 rose by 40%. PiP predicts the next ten years of work better than the h-index in every setting and in each of 15 fields (at career age 10, Spearman 0.43 against 0.35 for the number of future top-10% papers); the lead narrows but remains when eligibility does not depend on citations. Other indicators built from the same paper percentiles each beat PiP on at most one of three outcomes; compared with a sum of percentiles above the same cutoff, the peer comparison at every threshold mainly improves the prediction of future citation impact. PiP reaches the h-index's age-10 accuracy in finding future high performers 1.6 to 1.9 years sooner, and anticipates later academy memberships and fellowships better than the h-index (AUC 0.75 against 0.71 at career age 5), though only slightly within already notable groups. The results hold without self-citations and agree with Google Scholar profiles ($\rho = 0.78$).

Keywords: research evaluation; citation percentiles; h-index; career age; indicators; Semantic Scholar

1 Introduction

Hiring, promotion and award committees compare scholars with others at a similar stage of their careers. A committee deciding on tenure asks whether a candidate's record is strong for someone six years past the doctorate; a funding panel compares early-career applicants across departments. Bibliometric indicators enter these decisions alongside expert judgment, and the most common ones, citation counts and the h-index (Hirsch, 2005), are hard to read in this setting. Their values depend on the field, grow with seniority and drift with the calendar: the average h-index of a scholar ten years into their career is 40% higher for those who

[*]Department of Technology, Operations, and Statistics, Stern School of Business, New York University.
[†]Corresponding author: panos@stern.nyu.edu

started in 2010 than for those who started in 1960 (Section 5.3). A raw value says little until it is compared with the values of comparable scholars.

Many of these decisions are made early. Hiring and tenure committees, fellowship panels and early-career grants judge scholars three to eight years into their careers, when most of their papers are recent and few have had time to collect many citations. An indicator is most useful in this setting if it detects strong records early: if, measured at year five, it already separates the scholars who will go on to do important work from those who will not. Detecting performance early was one of the motivations for this work.

The h-index also reads only part of a record. It is the largest h such that h papers have at least h citations each: the corner where the sorted citation curve crosses the diagonal. Papers below the corner do not count, the citations of papers above it beyond h do not count, and a recent paper counts only once it has collected h citations, however well it compares with papers of its own age. A scholar with one landmark paper and a scholar with one average paper both have $h = 1$.

At the level of single papers, the field has settled on percentiles as the preferred normalization (Bornmann et al., 2013; Waltman, 2016; Hicks et al., 2015): a paper is compared with papers of the same publication year, and possibly the same field, by the share of them it out-cites. Percentiles do not depend on the skewed tail of the citation distribution, they need no assumption about how citations grow with age, and their meaning does not change with the calendar. Combining paper percentiles into an indicator for a scholar is less settled. Sums of percentiles, such as I3 (Leydesdorff and Bornmann, 2011), grow without bound with the number of papers, so many weak papers can outweigh one strong paper. Means ignore volume altogether. Counts of papers in the top 10% (Waltman et al., 2012; Bornmann and Marx, 2014) use one threshold and discard everything below it.

This paper proposes PiP, short for *percentile in percentile*, which places a scholar's papers in percentiles of their publication year and then places the scholar in percentiles of their career cohort. For each impact threshold θ, we count the scholar's papers at or above percentile θ and ask what share of peers, scholars who published their first paper in the same year, have fewer such papers. This standing, traced over all thresholds, forms a curve (Figure 1). PiP is the average height of the curve above a cutoff τ. Where the h-index reads one corner of a record, PiP measures the area under the whole curve. With $\tau = 0.5$, which we use throughout, only papers in the top half of their year can earn credit, recent papers can earn it as soon as they stand out among papers of their own age, and a single paper in the top 2.5% counts at every threshold of the computation, including the highest ones, where most peers have nothing. Lowering τ moves weight towards the number of papers; raising it moves weight towards their impact.

We compute PiP from the Semantic Scholar Academic Graph (Kinney et al., 2023) for 11.1 million authors at every career age from 1 to 30, reconstructing peers and percentiles in each evaluation year from the publications and citations up to that year. The paper makes six contributions:

1. **Definition and properties.** PiP is bounded, never decreases when a paper is added or gains citations, gives nothing to papers below the cutoff, and reads as the probability that the scholar has more papers than a random peer above a random impact threshold (Section 4).

2. **Measurement over time.** Paper percentiles settle within three to five years, with field-specific windows (Section 5.1); the distribution of PiP changes little across career cohorts from 1960 to 2015 (Section 5.3);

and a scholar's PiP is less stable over a career than their article count, more stable than their h-index, and more stable than their citations over horizons of ten years or more (Section 5.4).

3. **Prediction.** PiP predicts a scholar's future top papers better than the h-index in every setting and field we test, whether the h-index is used raw or converted to a cohort percentile; with eligibility that does not depend on citations, the lead narrows but remains. Against counts of top papers and sums and means of paper percentiles it offers a balance rather than dominance: none of them beats it on more than one of three outcomes. A sum of percentiles above the same cutoff predicts the number of future top papers about as well; the comparison with peers at every threshold adds to predicting the citation impact of future work and the chance of an exceptional paper (Section 5.5).

4. **Early detection.** Measured at the same career age, PiP identifies future high performers at least as accurately as the other indicators we test, with the sum of percentiles above the cutoff and the original PiP-AUC close behind, and it reaches the accuracy of the h-index at career age 10 1.6 to 1.9 years sooner (Section 5.6).

5. **Later recognition.** Measured early in a career, PiP anticipates later academy memberships, fellowships and prizes better than the h-index among all scholars; within groups that are already notable the lead is small. For this purpose the impact end of the dial works best (Section 5.7).

6. **Robustness.** The results hold without self-citations; PiP rankings from Semantic Scholar agree with rankings from author-curated Google Scholar profiles; and a field-normalized version removes most of the advantage of high-citation fields while largely preserving the order of scholars within a field (Sections 5.8–5.9).

An earlier version of this work (Tian and Ipeirotis, 2021) studied the predictability of paper and scholar percentiles on a smaller dataset of faculty at top US universities and rated scholars by the sum of their paper percentiles. Reviewers of that version objected to the sum, which rewards volume without bound, to the selected population, and to the lack of field-level results. This paper replaces the sum with PiP, moves to a population of all authors who meet an eligibility rule at each evaluation year, including those who later stop publishing, and reports results by field. A first, asymmetric form of the score was described by Ipeirotis (2024) and is used by the website `https://www.pip-score.org`; Section 4 explains why we replace it.

2 Related work

The h-index and its variants. The h-index (Hirsch, 2005) combines productivity and impact in one number and has been studied, criticized and modified at length. The g-index (Egghe, 2006) gives credit to the citations above the corner; the m-quotient (Hirsch, 2005), the contemporary h-index (Sidiropoulos et al., 2007) and hI_a (Harzing et al., 2014) correct for career length by dividing by academic age or discounting old papers; Kaur et al. (2013) rescale the h-index by the field average. Most variants correlate highly with the original (Bornmann et al., 2011). They keep its basic construction, a count of papers above a citation threshold that

depends on the scholar, and so keep its blindness to papers outside the core. The h-index is also inconsistent: adding the same paper to two scholars can reverse their order (Waltman and van Eck, 2012).

Percentiles. Percentile-based indicators compare each paper with a reference set of papers from the same year and field (Bornmann et al., 2013; Leydesdorff et al., 2011; Waltman and Schreiber, 2013; Bornmann and Williams, 2020). They are less sensitive to outliers than mean-based normalization (Schubert and Braun, 1986; Radicchi et al., 2008; Waltman and van Eck, 2019) and are recommended in the Leiden Manifesto (Hicks et al., 2015). Ties and the definition of the percentile need care (Waltman and Schreiber, 2013); we use the share of the reference set with strictly fewer citations for scoring, so that uncited papers have percentile zero, and Hazen's mid-rank (Hazen, 1914) for correlations (Section 4.4 discusses the choice). At the level of a scholar, Bornmann and Marx (2014) propose counting papers in percentile classes and normalizing by academic age, and I3 (Leydesdorff and Bornmann, 2011; Rousseau, 2012) sums percentiles over a set of papers. Composite indicators such as the one of Ioannidis et al. (2019) combine several citation counts, and Abramo and D'Angelo (2014) weight field-normalized citations by inputs. For countries and institutions, double rank analysis gives each of a unit's papers a rank within the unit and a rank among all papers in the world, and reads the unit's output at very high percentiles from the relation between the two (Rodríguez-Navarro and Brito, 2018; Brito and Rodríguez-Navarro, 2018). PiP differs from these in normalizing twice: papers by their publication year, and the scholar by their career cohort, at every impact threshold. The cohort comparison is made threshold by threshold: the count of a scholar's papers above each percentile is ranked among career peers, rather than read against the world distribution of papers.

Prediction of scholarly impact. Hirsch (2007) found that the h-index predicts a scholar's later h-index and citations better than counts of papers or citations do. Later work predicted future h-index values from features of a career (Acuna et al., 2012), questioned how much of that predictability is mechanical, since the future h-index contains the current one (Penner et al., 2013), and modeled impact as a stable individual parameter applied to papers of random quality (Sinatra et al., 2016); Clauset et al. (2017) review the area. We avoid the mechanical overlap by predicting outcomes built only from papers published after the evaluation year.

Citation windows and cumulative advantage. How long a paper needs before its citations reflect its impact depends on the field (Wang, 2013; Abramo et al., 2011). Citations also feed on themselves: well-known scholars receive more attention for the same work (Merton, 1968; Petersen et al., 2014), which affects any citation-based indicator and any test of its predictive validity. We return to this in Section 6.

3 Data

Papers and citations. We use the Semantic Scholar Academic Graph (Kinney et al., 2023), release of 27 October 2026, which is distributed under the ODC-BY license. It contains 226.0 million papers published from 1950 to 2025. For every paper we count citations by the year of the citing paper, which gives the cumulative citations at each age t (the number of years after publication), through calendar year 2025. These

counts add up to 97.5% of the citation counts that Semantic Scholar reports for papers from 2000–2005, with a correlation of 0.999; the difference consists of citations from 2026 and from citing papers without a year.

Articles. Semantic Scholar contains many records that are not research articles. We keep the 71.5 million papers typed as journal articles, conference papers, reviews, studies, clinical trials, meta-analyses or case reports, and not as editorials, news or letters. Only these *articles* enter the reference sets, the paper counts and the h-index. In the 2000 cohort, 63% of all records are uncited at age 5, but only 30% of articles. Each article carries zero or more of Semantic Scholar's 23 fields of study.

Authors and careers. Semantic Scholar assigns author identifiers through automatic disambiguation. We link authors to their articles (277.8 million links) and define an author's *start year* s_a as the year of their first article, and their *career age* at evaluation year Y as $A = Y - s_a$. We follow every author from career age 1 to 30, or to 2025, whichever comes first. At each evaluation year an author is *eligible* if they have at least three articles and an h-index of at least three, both computed from citations received through Y. Because both counts only grow, an author who becomes eligible remains eligible, also after they stop publishing. In total, 11.1 million authors are eligible at some career age, giving 111.6 million author-age records. The h-index condition is strict early in a career and in fields that cite slowly: among authors with at least three articles at career age 3 (start years 1975–2000), it excludes 79%, and at career age 10 it excludes 43%, from 22% in biology to 64% in economics and 97% in history. Sections 5.6 and 5.9 repeat the main comparisons with eligibility defined by the three articles alone.

Every quantity used at evaluation year Y is computed from the articles published up to Y and the citations they received up to Y. The reconstruction is retrospective in one respect: author identities, publication types and fields come from the 2026 release, not from the records available at the time.

Google Scholar profiles. For validation we use 16,218 Google Scholar profiles collected for `https://www.pip-score.org`, mostly of faculty. Unlike Semantic Scholar's author records, these profiles are curated by the scholars themselves, so they are a check on automatic disambiguation (Section 5.9).

4 The PiP score

4.1 Paper percentiles

Fix an evaluation year Y. For an article j published in year $p_j \le Y$, let $c_j(Y)$ be the citations it has received through Y. Its *percentile* is the share of articles published in the same year with strictly fewer citations at Y:

$$y_j(Y) = \frac{\left|\{\, i : p_i = p_j,\ c_i(Y) < c_j(Y) \,\}\right|}{\left|\{\, i : p_i = p_j \,\}\right|} \in [0,1). \tag{1}$$

Uncited articles have $y_j = 0$. Every article is compared only with articles of the same publication year, evaluated at the same time, so y_j needs no correction for the age of the paper or for the growth of citations

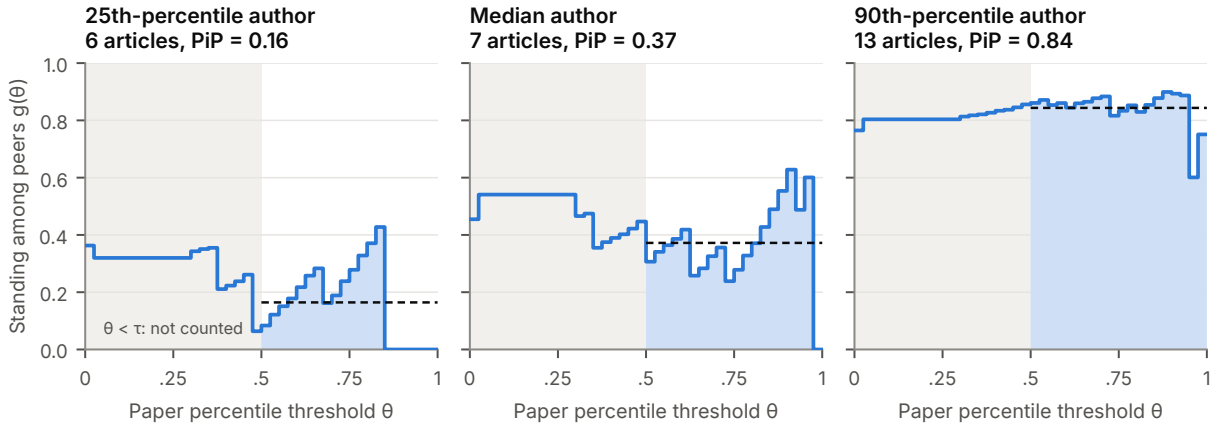


Figure 1: Standing curves of three authors who published their first article in 1995, evaluated in 2005 (career age 10). Each was chosen as the author whose PiP is closest to the 25th, 50th or 90th percentile of the cohort's scores. The curve gives, for each threshold θ, the share of the 1995 cohort's eligible authors who have fewer articles at or above percentile θ of their publication year. PiP is the shaded area to the right of $\tau = 0.5$, rescaled by $1/(1-\tau)$, or equivalently the average height of the curve there (dashed line). The median author has seven articles, two of them in the top 10% of their year and one in the top 5%. At thresholds from 0.85 to 0.95 this author stands above about half of the peers, who have fewer articles that highly cited.

over the decades. A paper one year old competes only with other papers one year old. The main analysis pools all fields; Section 5.8 computes percentiles within fields.

4.2 Peers and standing curves

At evaluation year $Y = s_a + A$, the *peers* $\mathscr{P}$ of author a are the authors with the same start year who are eligible at Y, the author included. For an impact threshold $\theta \in [0,1]$, let

$$n_a(\theta) = \left|\{\, j \in a : p_j \le Y,\ y_j(Y) \ge \theta \,\}\right| \tag{2}$$

be the number of the author's articles at or above percentile θ; $n_a(0)$ is the number of articles. The author's *standing* at threshold θ is the share of peers with strictly fewer articles at or above θ:

$$g_a(\theta) = G_\theta(n_a(\theta)), \qquad G_\theta(m) = \frac{\left|\{\, b \in \mathscr{P} : n_b(\theta) < m \,\}\right|}{|\mathscr{P}|}. \tag{3}$$

An author with no article at or above θ stands at zero there, however many peers share that count (Section 4.4 discusses ties). The *standing curve* $\theta \mapsto g_a(\theta)$ starts at the author's standing on the number of articles and ends at their standing on the number of top articles. It need not be monotone: an author can rank higher among peers on the number of top-10% papers than on the number of papers (Figure 1).

4.3 Definition

The PiP score with cutoff $\tau \in [0,1)$ is the average standing over the thresholds at or above τ:

$$\mathrm{PiP}_\tau(a) = \frac{1}{1-\tau}\int_\tau^1 g_a(\theta)\,d\theta. \tag{4}$$

It is the area under the standing curve to the right of τ, rescaled to $[0,1]$. We use $\tau = 1/2$ throughout, so only articles in the top half of their publication year can earn credit, and write PiP for $\mathrm{PiP}_{1/2}$. Section 5.2 shows how the cutoff moves the score between quantity and impact.

The score has a direct probabilistic reading. If B is a peer drawn at random and Θ a threshold drawn uniformly from $[\tau,1]$, then

$$\mathrm{PiP}_\tau(a) = \Pr\left[\, n_a(\Theta) > n_B(\Theta) \,\right]: \tag{5}$$

PiP is the probability that the author has more articles than a random peer above a random impact threshold. Averaging over the peers gives the mean score of a peer group. Let $p_{\theta,m}$ be the share of peers with exactly m articles at or above θ. Two random peers tie at θ with probability $\sum_m p_{\theta,m}^2$ and are otherwise equally likely to be ahead, so

$$\frac{1}{|\mathscr{P}|}\sum_{a\in\mathscr{P}} \mathrm{PiP}_\tau(a) = \frac{1}{2(1-\tau)}\int_\tau^1 \Big(1-\sum_m p_{\theta,m}^2\Big)\,d\theta. \tag{6}$$

The mean score of a cohort is below one half and falls as ties become more common, which matters at high cutoffs (Section 5.2).

4.4 Properties

The following hold for a fixed peer group that contains the author, both for the integral and for the computed score of Section 4.6; each follows in a line from the definitions. We also checked them numerically on simulated cohorts, on the grid of the computation and on a fine grid (the code is in the replication package).

Range. $0 \le \mathrm{PiP}_\tau(a) < 1$. The author is one of their own peers and does not have fewer articles than themselves, so $g_a(\theta) \le 1 - 1/|\mathscr{P}|$ at every threshold. The score is zero when, at almost every threshold above τ, no peer has fewer articles at or above that threshold than a, for example when a has no article above τ.

Monotonicity. Adding an article, or raising the percentile of an existing article, never lowers PiP_τ, because either change weakly increases $n_a(\theta)$ at every threshold and G_θ is non-decreasing. The gain can be zero: one more article at or above θ raises the standing at θ only if some peer has exactly as many such articles as the author had before. An article whose percentile stays below τ changes $n_a(\theta)$ only for $\theta < \tau$ and leaves the score unchanged.

No reward for volume alone. At $\tau = 1/2$, an author whose articles all fall below the median of their publication year scores zero, however many such articles they publish.

Credit for the best work. A single article in the top 2.5% of its year raises $n_a(\theta)$ by one at every threshold of the computation, including the highest, where most peers have none. In a simulated cohort of 2,000 peers, a record of one such article scores 0.45, while one article at the 50th percentile scores 0.004. With the exact integral, one article at the 97.5th, 99th or 99.9th percentile scores 0.42, 0.45 or 0.47, and one at the 50th percentile scores 0. The h-index cannot tell any of these records apart (all have $h = 1$).

Ties. The standing counts peers with strictly fewer articles. Mid-ranks, which count half of the tied peers, are a common choice for the percentile of a single paper (Waltman and Schreiber, 2013), but here they would give credit to authors with nothing above a threshold: at high thresholds most peers have no article, and an author tied with them at zero would stand at half their share. In the simulated cohort, an author whose five articles all fall below the median would score 0.22 with mid-ranks instead of 0. The strict rule keeps weak papers from earning credit through ties; its price is that the mean score of a cohort depends on how common ties are (Eq. 6).

Factorization. Let $F(m) = |\{b \in \mathscr{P} : n_b(0) \le m\}|/|\mathscr{P}|$ be the productivity percentile. Unlike G_θ it counts tied peers, so that it is positive for every member of the peer group. Defining the *impact multiplier* $Q_\tau(a) = \mathrm{PiP}_\tau(a)/F(n_a(0))$ gives

$$\mathrm{PiP}_\tau(a) = F(n_a(0)) \cdot Q_\tau(a). \tag{7}$$

The identity holds by the definition of Q_τ; it describes a score rather than explaining it. The multiplier compares the author's standing on counts of articles above the cutoff with their standing on the number of articles, and it exceeds one when the author ranks higher among peers on the former. Because F counts ties and G_θ does not, an author whose standing is the same at every threshold has Q_τ slightly below one. In our data $Q_{1/2}$ exceeds one for 30–36% of eligible authors, depending on career age. Across eligible authors who started in 1975–2005, at career ages 5, 10 and 20, the variance of $\log \mathrm{PiP}_{1/2}$ splits into 0.74–0.82 from $\log Q$, about 0.30 from $\log F$ and a small negative covariance (-0.04 to -0.11). These shares describe how the scores vary; they do not measure how much of PiP reflects research quality.

Consistency. PiP is not consistent in the sense of Waltman and van Eck (2012): adding the same article to two authors can reverse their order, because G_θ is not linear. In simulation, this happened in 0.7% of 3,000 random pairs. The h-index shares the property.

4.5 Relation to other indicators

Writing indicators in terms of the count curve $\theta \mapsto n_a(\theta)$ places PiP among existing ones. A top-x% count reads the curve at one threshold, $n_a(1-x)$, without comparing it with peers. I3 sums the percentiles, $\sum_j y_j = \int_0^1 n_a(\theta)\, d\theta$: the same integral without the peer comparison, so it grows without bound with the number of articles. Restricting I3 to the cutoff gives the *cutoff-matched sum*

$$T_\tau(a) = \frac{1}{1-\tau} \sum_j \max(y_j - \tau, 0) = \frac{1}{1-\tau} \int_\tau^1 n_a(\theta)\, d\theta, \tag{8}$$

which uses the same percentiles and the same cutoff as PiP_τ and differs from it only in the comparison with peers inside the integral; Section 5.5 uses it to measure what that comparison adds. The h-index reads one point of the analogous curve in citation space, where the number of articles with at least k citations equals k.

The first form of the score, which we call the *original PiP-AUC* (Ipeirotis, 2024), sorts an author's articles by percentile, $y_{(1)} \geq y_{(2)} \geq \cdots$, and computes the area under the step curve that plots $y_{(k)}$ against the productivity percentile $F(k)$:

$$\text{PiP-AUC}(a) = \sum_{k=1}^{n_a(0)} y_{(k)} \left[F(k) - F(k-1)\right] = \int_0^1 F(n_a(\theta))\, d\theta. \tag{9}$$

The second expression shows the difference from PiP: the original form compares the author's count at or above θ with peers' *total* counts rather than with their counts at or above θ. It is the product of $F(n_a(0))$ and a weighted mean of the author's own percentiles, which keeps it bounded and monotone. But the weight of the k-th best article, $F(k) - F(k-1)$, is the share of peers with exactly k articles, which is zero for every k below the smallest article count among peers. Among the eligible authors of a cohort, nobody has fewer than three articles, so an author's best two articles receive no weight at all; among Google Scholar faculty profiles, the median author's best four articles receive none. The original form is therefore dominated by productivity: the productivity percentile accounts for 0.59–0.77 of the variance of its logarithm, and an author with one article at the 99th percentile scores zero, while an author with 40 weak articles scores 0.10 in the simulated cohort. A cutoff τ applied to the original form compares partial counts with growing total counts, and its median at career age 10 halves between the 1960 and 2010 cohorts. The symmetric comparison in PiP removes both problems.

4.6 Computation

We evaluate the integral on the grid $\theta_k = k/40$, $k = 0, \ldots, 39$, so that $\mathrm{PiP}_{1/2}(a) = \frac{1}{20} \sum_{k=20}^{39} g_a(\theta_k)$. On the interval $(k/40, (k+1)/40]$ an article counts exactly when $y_j \geq k/40$, so this sum equals the integral of Eq. 4 with every percentile rounded up to the next multiple of $1/40$. The reported score is therefore PiP on 40 percentile classes, and the properties above hold for it as stated. The classes have two visible effects: articles in the top 2.5% of their year are not told apart, and an article in the class just above the median earns a full class of credit.

To check that the classes do not drive the results, we also computed the exact integral for the 5.3 million eligible authors at career age 10 (start years 1960–2015). On $(y_{(k+1)}, y_{(k)}]$, where $y_{(k)}$ is the author's k-th best percentile, the author has k articles at or above θ, and a peer has fewer exactly when its own k-th best percentile is below θ, so the integral reduces to sorting the peers' k-th best percentiles. Against the exact score, the 40-class score is 0.010 higher on average at $\tau = 0.5$ (0.055 at most) and ranks authors within cohorts almost identically: the rank correlation is 0.9994, and 96% of each cohort's top 10% is the same (start years 1975–2010). At $\tau = 0.75$ and 0.9 the correlation is 0.998 and 0.996 and the overlap 93% and 85%. None of the prediction results at career age 10 (Section 5.5) changes by more than 0.004 at any of the three cutoffs, so we keep the 40 classes. For each start year and evaluation year, the distributions G_{θ_k} come from one pass over the peers' counts. The whole computation, 111.6 million author-age records, runs as a set

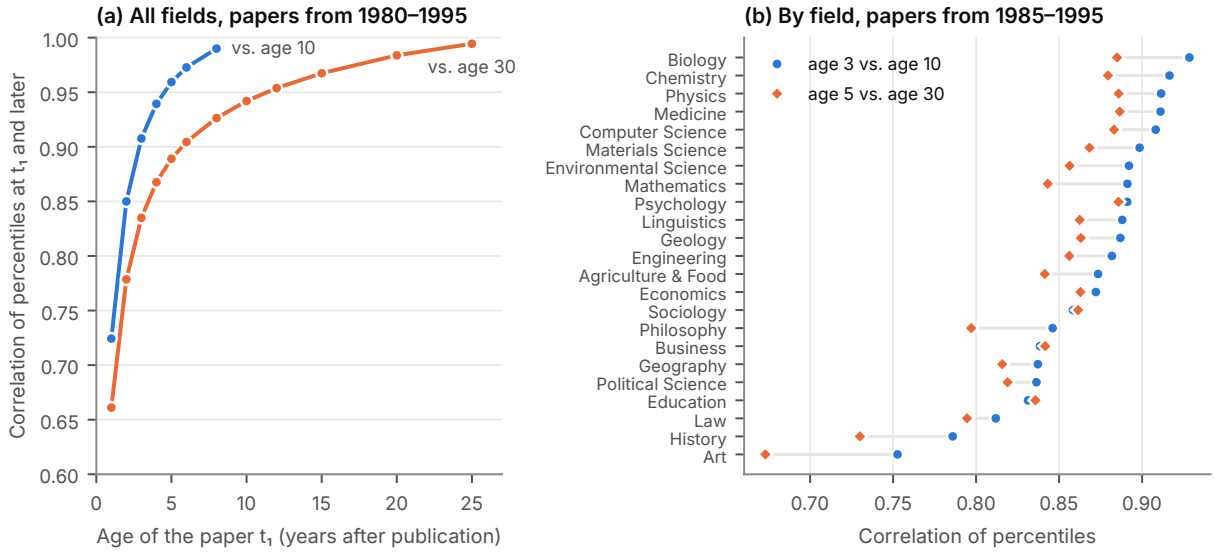


Figure 2: How well a paper's early citation percentile anticipates its later percentile. (a) Correlation between the percentile at age t_1 and at age 10 or 30, for 7.2 million articles published in 1980–1995, averaged over publication years. (b) The same correlation by field, for articles published in 1985–1995, with percentiles computed within the field and publication year; fields with more than 20,000 articles.

of SQL queries in BigQuery.

5 Results

5.1 Paper percentiles settle within a few years

A scholar's PiP uses percentiles of recent papers, measured after only a few years of citations. We first check how well an early percentile anticipates the later one. For every article published from 1980 to 1995, we compute its percentile within its publication year at each age, using Hazen mid-ranks (Hazen, 1914), and correlate the percentiles at two ages within each publication year. Because the percentiles are ranks, this is the Spearman correlation of the citation counts. We report the average over publication years.

Percentiles settle quickly (Figure 2a, Table 1). The correlation with the percentile at age 10 is 0.72 after one year, 0.91 after three and 0.96 after five; with the percentile at age 30 it is 0.84 after three years and 0.89 after five. The ordinary least-squares slope of the later percentile on the earlier one is close to one (0.94 for ages 3 and 10), so early percentiles are not strongly biased towards the middle. High correlations still leave room for individual movement: between ages 3 and 10 a paper's percentile changes by 0.086 on average (Table 1). Early percentiles have become more informative over time: the correlation between ages 1 and 10 was 0.72 for publication years before 1995 and 0.83 for 2015. For each cohort from 1960 to 1995, the correlation between ages 3 and 10 lies between 0.875 and 0.910.

The speed differs by field (Figure 2b), in line with Wang (2013) and Abramo et al. (2011). Three years give a correlation of about 0.9 with age 10 in biology (0.93), chemistry, physics, medicine and computer science (0.91), but 0.79 in history and 0.75 in art. The humanities and law need about five years to reach the

Table 1: Agreement between a paper's citation percentile at an early and a later age (articles published 1980–1995, 7.2 million papers). Correlation of Hazen percentiles within publication year, the slope of the later on the earlier percentile, and the mean absolute change.

Ages	Correlation	Slope	Mean absolute change
1 vs. 10	0.724	0.846	0.161
2 vs. 10	0.850	0.911	0.115
3 vs. 10	0.908	0.944	0.086
5 vs. 10	0.959	0.975	0.051
3 vs. 30	0.835	0.882	0.125
5 vs. 30	0.889	0.918	0.100
10 vs. 30	0.942	0.956	0.068
20 vs. 30	0.984	0.989	0.030

Table 2: PiP at different cutoffs τ, career age 10. Columns 2–4: within-cohort Spearman correlation with characteristics of the record (3.43 million authors, start years 1975–2010). Columns 5–7: prediction of the next ten years (1.45 million authors, start years 1975–2000). Last column: median score of the 1960 and 2010 cohorts.

	Correlation with			Prediction			
τ	Articles	Mean pct.	Top 2.5%	Top-10% count	Top-1% AUC	Mean pct.	Median
0	0.856	0.214	0.491	**0.445**	0.766	0.223	0.44 → 0.40
0.25	0.776	0.329	0.537	0.443	0.770	0.265	0.43 → 0.39
0.5	0.637	0.466	0.602	0.427	**0.772**	0.321	0.41 → 0.38
0.6	0.573	0.506	0.633	0.415	0.771	0.340	0.41 → 0.37
0.75	0.473	**0.536**	0.688	0.388	0.766	**0.356**	0.38 → 0.33
0.9	0.370	0.486	**0.781**	0.347	0.750	0.338	0.37 → 0.16
Original PiP-AUC	0.897	0.131	0.421	0.434	0.756	0.190	0.29 → 0.26
h-index	–	–	–	0.352	0.719	0.230	–

agreement that the natural and life sciences reach in three. PiP uses percentiles at every age, including recent papers; Section 6 discusses what this means for scholars in slow fields.

5.2 The cutoff τ is a dial between quantity and impact

The cutoff decides which papers earn credit. We computed PiP for $\tau \in \{0, 0.25, 0.5, 0.6, 0.75, 0.9\}$ for 3.43 million eligible authors at career age 10 (start years 1975–2010) and correlated each version, within cohort, with the author's number of articles, the mean percentile of their articles and their number of articles in the top 2.5% of their year (Figure 3a, Table 2).

As τ rises, the correlation with the number of articles falls from 0.86 to 0.37 and the correlation with the number of top-2.5% articles rises from 0.49 to 0.78. The correlation with the mean percentile peaks at $\tau = 0.75$. The original PiP-AUC sits at the quantity end of this range, closer to the article count (0.90) than even $\tau = 0$. In prediction, the same dial trades the *number* of future top-10% papers, best predicted at low τ, for their *citation impact*, best predicted at $\tau = 0.75$; the chance of a future top-1% paper is predicted about equally well from $\tau = 0.25$ to 0.6. Above $\tau = 0.6$ the distribution of the score drifts across cohorts. The drift is not an effect of the 40 percentile classes (Section 4.6): with the exact integral, the median at $\tau = 0.9$ still

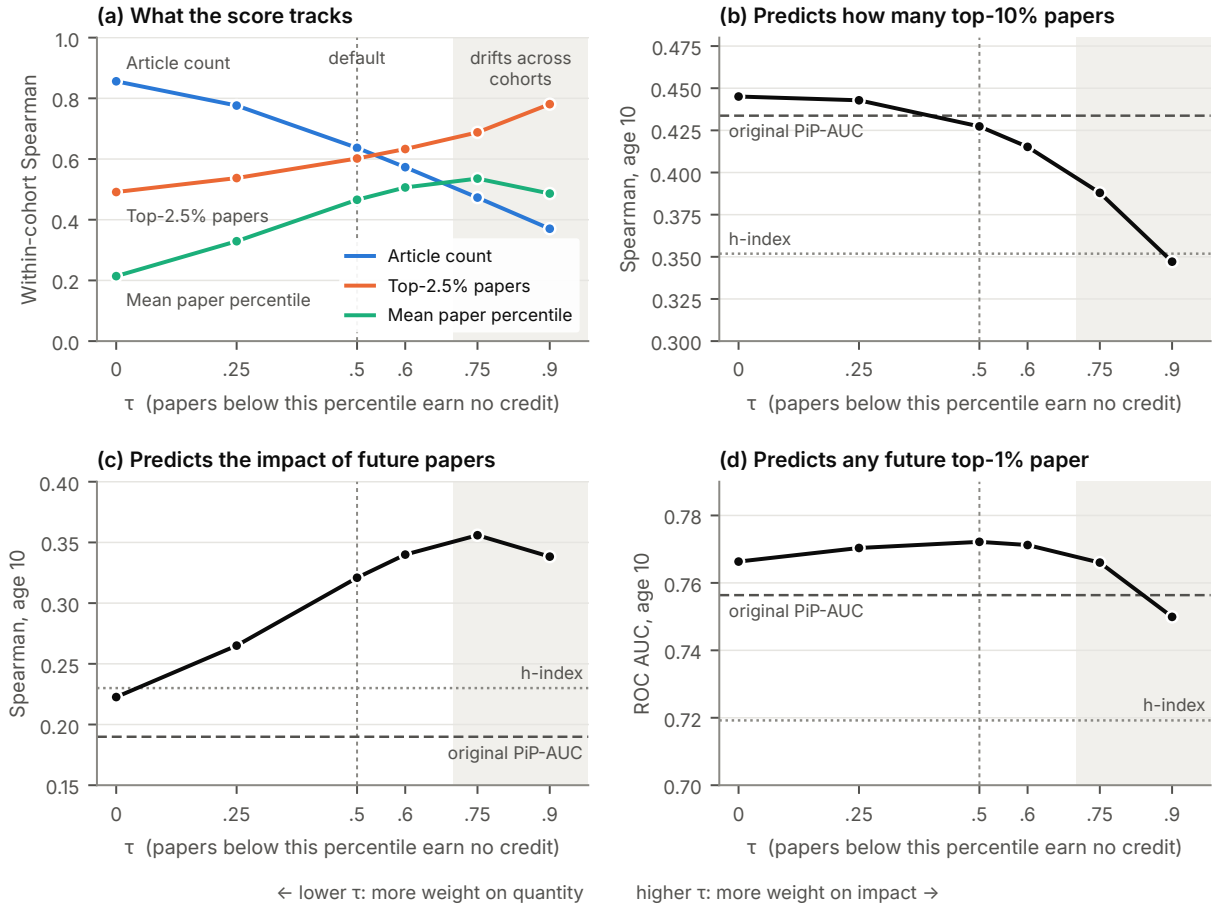


Figure 3: The cutoff τ moves PiP from quantity towards impact. Eligible authors at career age 10, start years 1975–2010. (a) Within-cohort Spearman correlation of PiP with the author's number of articles, number of articles in the top 2.5% of their year, and mean article percentile. (b–d) Prediction of the author's articles published in the next ten years (Section 5.5): (b) Spearman correlation with the number of future top-10% articles, (c) Spearman correlation with their mean percentile, (d) ROC AUC for publishing at least one top-1% article. Horizontal lines show the original PiP-AUC and the *h*-index. In the shaded range the score's distribution drifts across cohorts.

falls from 0.32 in the 1960 cohort to 0.12 in the 2010 cohort. It comes from ties at zero (Eq. 6): the share of eligible authors with no article in the top 10% of its year rises from 6% in the 1960 cohort to 42% in the 2010 cohort, and all of them score zero at $\tau = 0.9$.

We use $\tau = 0.5$ as the default. It has a one-line explanation (only papers in the top half of their year count), it balances the correlations with quantity (0.64) and with top papers (0.60), it is at the maximum for the top-1% outcome, and it is stable across cohorts. Users who care more about volume can lower it, and users who care more about impact can raise it up to about 0.6; the choice should be made before the evaluation and stated.

Retrospective temporal robustness check. The default was chosen from results that pooled all cohorts, so we checked afterwards whether the choice depends on them. Because the check was made after the default was chosen, it does not protect against specification search as an untouched holdout would. We split the

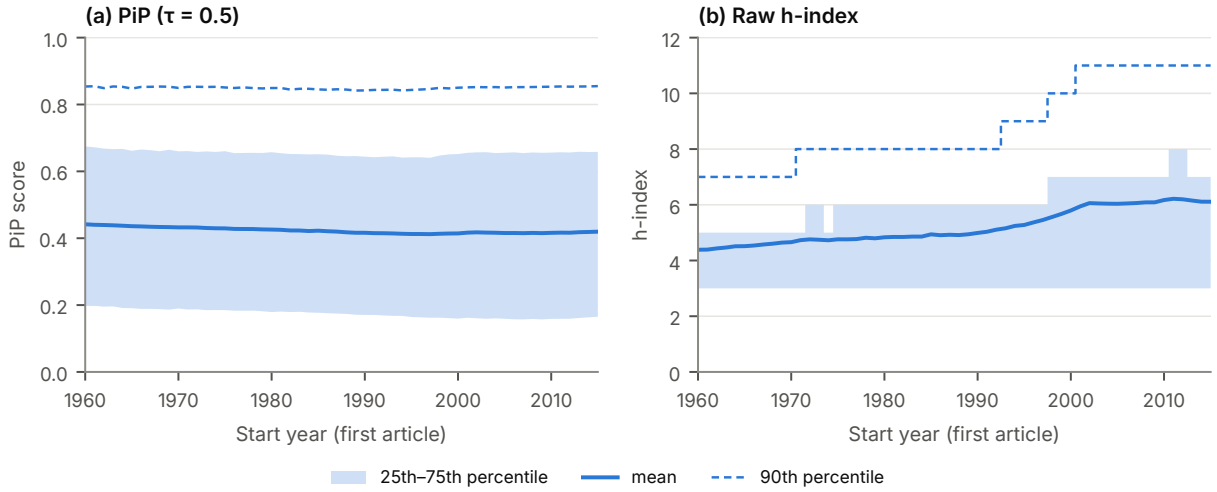


Figure 4: Distribution of (a) PiP ($\tau = 0.5$) and (b) the raw h-index of eligible authors at career age 10, by start year. The PiP distribution changes little over 55 cohorts; the mean h-index grows by 40%.

same-age prediction settings by start year into a choice sample (1975–1990) and a test sample (1991 and later). Applying the rule above to the choice sample alone (the highest AUC for a future top-1% paper among cutoffs whose distribution is stable across cohorts, $\tau \leq 0.6$) selects $\tau = 0.6$ at career age 10, with $\tau = 0.5$ within 0.001. In the test cohorts both cutoffs keep every comparison reported in Sections 5.5–5.6 at career ages 5, 10 and 15: they beat the h-index on all three outcomes, no other indicator beats either of them on more than one outcome, and both have the highest AUC for a future top-1% paper. Across τ from 0.25 to 0.6 that AUC varies by less than 0.01 in the test cohorts, so the exact choice in this range matters little for the outcome used to make it. Two differences between the samples are worth noting. The quantity–impact trade-off turns at a slightly higher cutoff in the older cohorts (between 0.6 and 0.75 rather than at 0.5), and in those cohorts at career age 5 the top-10% count ties $\tau = 0.5$ on the top-1% outcome and beats it on future citation impact, while $\tau = 0.6$ keeps its lead.

5.3 The distribution of PiP is stable across cohorts

Committees compare people who started in different years, and indicators are often read against fixed rules of thumb. Figure 4 shows the distribution of PiP and of the raw h-index at career age 10 for every start year from 1960 to 2015. The PiP distribution changes little: the mean goes from 0.44 to 0.42, the lower quartile from 0.20 to 0.16, the upper quartile stays at 0.66–0.67 and the 90th percentile at 0.85. Over the same cohorts the mean h-index at age 10 rises from 4.4 to 6.2 and its 90th percentile from 7 to 11, following the growth of publication and citation rates. The cohorts themselves grow nearly a hundredfold, from 4,031 eligible authors in 1960 to 392,360 in 2015.

Part of this stability is built in. Because the standing counts peers with strictly fewer articles, the mean score of a cohort is one half minus half the probability that two random peers tie at a random threshold above τ (Eq. 6), so it can move only as far as ties change. The rest of the distribution is not fixed by the construction, and where ties at zero become common it does drift: the original form with a cutoff drifts (Section 4), and so

Table 3: Stability of scholar indicators over a career. Correlation between an indicator's within-cohort percentile at two career ages (equal to the Spearman correlation of the raw values within cohort); eligible authors with start years 1975–1995. The 5-vs-30 comparison covers 434,000 authors.

Indicator	5 vs. 10	5 vs. 15	5 vs. 30	10 vs. 30	20 vs. 30
Articles	0.847	0.754	0.642	0.800	0.946
PiP ($\tau = 0.5$)	0.781	0.687	0.563	0.726	0.911
Original PiP-AUC	0.772	0.674	0.565	0.742	0.923
Citations	0.798	0.659	0.451	0.620	0.886
h-index	0.604	0.496	0.370	0.599	0.877

does PiP above $\tau = 0.6$ (Section 5.2). What the stability establishes is limited to rarity: a PiP of 0.6 at age 10 was about as rare in the 1970 cohort as in the 2010 cohort, so the scores of people at the same career age can be read as statements of rarity, whatever their start year. A cohort percentile of the h-index offers the same kind of scale. Stable rarity does not by itself mean that equal scores predict equal futures in different cohorts; Section 5.5 tests prediction when cohorts and seniorities are pooled.

5.4 Stability over a career

An indicator that reshuffles scholars every few years is of little use for decisions with long consequences. For authors who started in 1975–1995, Table 3 correlates each indicator's within-cohort percentile at two career ages. PiP is about as stable as the original PiP-AUC and much more stable than the h-index: between ages 5 and 30 its correlation is 0.56, against 0.37 for the h-index and 0.45 for citations. The article count is more stable at every horizon, as one would expect from a quantity that only accumulates; citations are slightly more stable than PiP between ages 5 and 10 and less stable over longer horizons. The h-index is particularly unstable early in a career, partly because many eligible authors share the values 3 and 4.

5.5 Predicting future work

The most demanding test of an indicator meant for career decisions is whether it anticipates what the scholar does next. For each eligible author at evaluation year Y, we take the articles they publish in years $Y+1$ to $Y+10$ and score each at its own age 5, within its publication year. No article counts towards both the predictor and the outcome. We use three outcomes: the number of future articles in the top 10% (Spearman correlation), whether at least one future article is in the top 1% (ROC AUC), and the mean percentile of the future articles among authors with at least one (Spearman correlation), which we call their citation impact.

Within one cohort, a raw indicator and its cohort percentile order people identically, so the comparison pools cohorts in two ways. *Same career age*: all eligible authors at career age $A = 5$, 10 or 15, with start years from 1975 to $2010 - A$, so that every article published in the ten years after the evaluation has five years of citations by 2025 (start years 1975–2000 at career age 10). This is the case of a tenure committee comparing candidates of similar seniority. *Same year*: all eligible authors in 2000 or in 2005, at career ages 2 to 30 (the case of a panel comparing people of different seniority). Pooling start years, a raw indicator and its cohort percentile no longer rank people identically, so in the second setting we compare each indicator raw

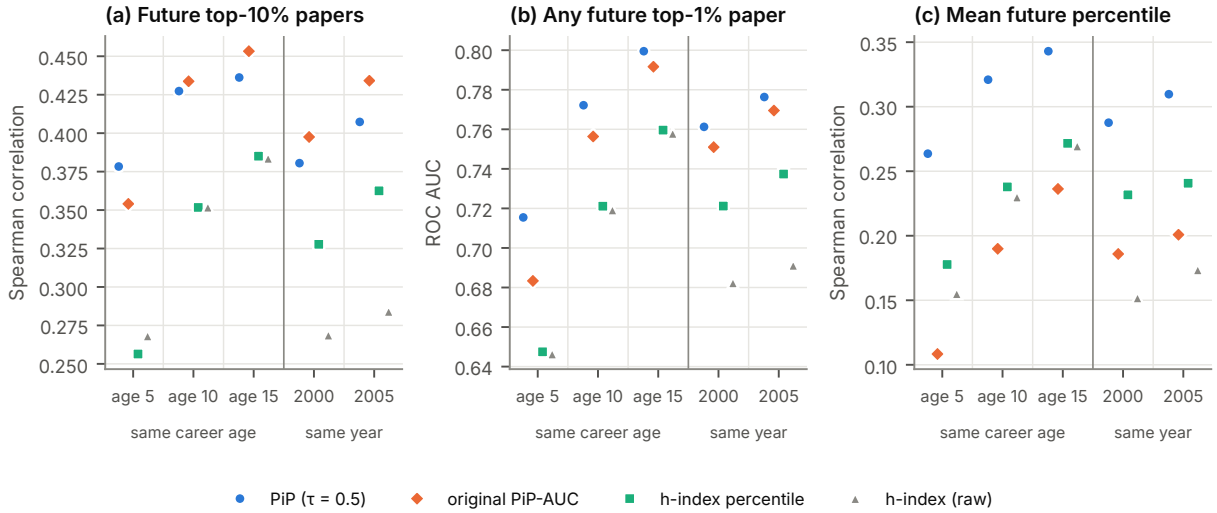


Figure 5: Prediction of the next ten years of a scholar's work from indicators at the evaluation year. Settings: eligible authors at career age 5, 10 or 15 (left of the vertical line), or all eligible authors in 2000 or 2005 at career ages 2–30 (right). Outcomes: (a) Spearman correlation with the number of future articles in the top 10% of their year; (b) ROC AUC for at least one future top-1% article; (c) Spearman correlation with the mean percentile of future articles. Each future article is scored at age 5.

and as a percentile within its cohort.

PiP predicts better than the *h*-index on all three outcomes, in all five settings (Figure 5, Table 4). At career age 10, its correlation with the number of future top-10% papers is 0.43 against 0.35, its AUC for a future top-1% paper is 0.77 against 0.72, and its correlation with the citation impact of future papers is 0.32 against 0.23. The gap is larger when people of different seniority are compared: in 2005 the raw *h*-index reaches 0.28, 0.69 and 0.17. Converting the *h*-index into a cohort percentile helps (0.36, 0.74, 0.24) but does not close the gap. PiP gains much less from the conversion: in 2005 its cohort percentile reaches 0.42, 0.78 and 0.32, against 0.41, 0.78 and 0.31 for the raw score, because the raw score is already relative to the cohort and its distribution does not depend on the start year (Section 5.3). We report cohort percentiles for every indicator in the same-year settings.

Productivity and citations. The number of articles predicts the number of future top-10% papers as well as PiP or slightly better (0.44 against 0.43 at age 10), because productive authors keep producing; the original PiP-AUC, which is close to a productivity percentile, does the same. Neither predicts the citation impact of future work (correlations of 0.00 and 0.19). Total citations predict future citation impact about as well as PiP (0.30 against 0.32) but are weaker on the number of future top papers (0.29).

Percentile-based indicators. The closest alternatives to PiP use the same paper percentiles but combine them differently: the number of papers in the top 10% (Waltman et al., 2012), also divided by career age (Bornmann and Marx, 2014); the number in the top 1%; I3, the sum of the percentiles (Leydesdorff and Bornmann, 2011); and their mean. Each is the best or nearly the best on one outcome and weak on another (Table 4). The mean percentile predicts the citation impact of future papers best (0.39 at age 10, against 0.32

Table 4: Prediction of the next ten years of work. Career age 10 (1.45 million authors, start years 1975–2000; raw values, pooled over start years) and all eligible authors in 2005 (1.85 million authors, career ages 2–30; within-cohort percentiles, except where noted). Spearman correlation with the number of future top-10% articles, ROC AUC for at least one future top-1% article, and Spearman correlation with the mean percentile of future articles. The best value in each column is in bold. $T_{0.5}$ is the sum of percentiles above the cutoff (Section 4); the last row averages the cohort percentiles of the article count and of the mean percentile.

	Career age 10, raw			Year 2005, cohort pct.		
Indicator	Top-10%	Top-1%	Mean pct.	Top-10%	Top-1%	Mean pct.
PiP ($\tau = 0.5$)	0.427	**0.772**	0.321	0.416	**0.781**	0.318
Original PiP-AUC	0.434	0.756	0.190	0.433	0.770	0.201
h-index	0.352	0.719	0.230	0.363	0.737	0.241
raw, in 2005				0.284	0.691	0.174
Citations	0.290	0.714	0.298	0.312	0.733	0.303
Articles	0.437	0.740	0.000	**0.439**	0.757	0.019
Top-10% count	0.360	0.748	0.338	0.354	0.762	0.324
per year of career, raw				0.379	0.771	0.345
Top-1% count	0.250	0.641	0.228	0.262	0.698	0.225
I3 (sum of percentiles)	**0.438**	0.755	0.182	0.434	0.769	0.189
$T_{0.5}$ (sum above the cutoff)	0.430	0.765	0.296	0.422	0.778	0.296
Mean percentile	−0.043	0.523	**0.394**	−0.042	0.526	**0.404**
Articles + mean pct.	0.331	0.717	0.323	0.325	0.726	0.333

for PiP), but says nothing about how many top papers will follow (−0.04) or whether one of them will be in the top 1% (AUC 0.52). I3 predicts the number of future top papers as well as the article count does (0.44) but not their citation impact (0.18). The top-10% count predicts future citation impact slightly better than PiP (0.34 against 0.32) and the number of future top papers clearly worse (0.36 against 0.43).

What the peer comparison adds. The cutoff-matched sum T_τ (Section 4) uses the same percentiles and the same cutoff as PiP and differs from it only in the comparison with peers at each threshold, so it measures what that comparison adds. At career age 10, $T_{0.5}$ reaches 0.430, 0.765 and 0.296 on the three outcomes, against 0.427, 0.772 and 0.321 for PiP. Cohort by cohort the pattern is consistent: in all 26 start-year cohorts at career age 10, PiP is ahead of $T_{0.5}$ on the top-1% outcome (by 0.004 to 0.011) and on future citation impact (by 0.023 to 0.036), while on the number of future top-10% papers the two are within 0.01 of each other (PiP ahead in 16 cohorts). At career ages 5 and 15, PiP is again ahead on the first two outcomes in every cohort, and on the number of future top papers it is ahead in 29 of 31 cohorts at age 5 and in 6 of 21 at age 15. The comparison with peers thus adds little to predicting how many top papers will follow, which the sum above the cutoff already captures, and adds consistently to predicting how highly cited the future papers will be and whether one of them will be exceptional. A direct mix of quantity and impact does worse: the average of the cohort percentiles of the article count and of the mean percentile matches PiP on future citation impact (0.323) and falls far behind on the other two outcomes (0.331 and 0.717); weights of 0.25 or 0.75 on the article count move it towards the mean percentile or towards the article count without reaching PiP's balance.

In all five settings, each of these indicators, including T_τ at every cutoff from 0.25 to 0.9 and the mixes of

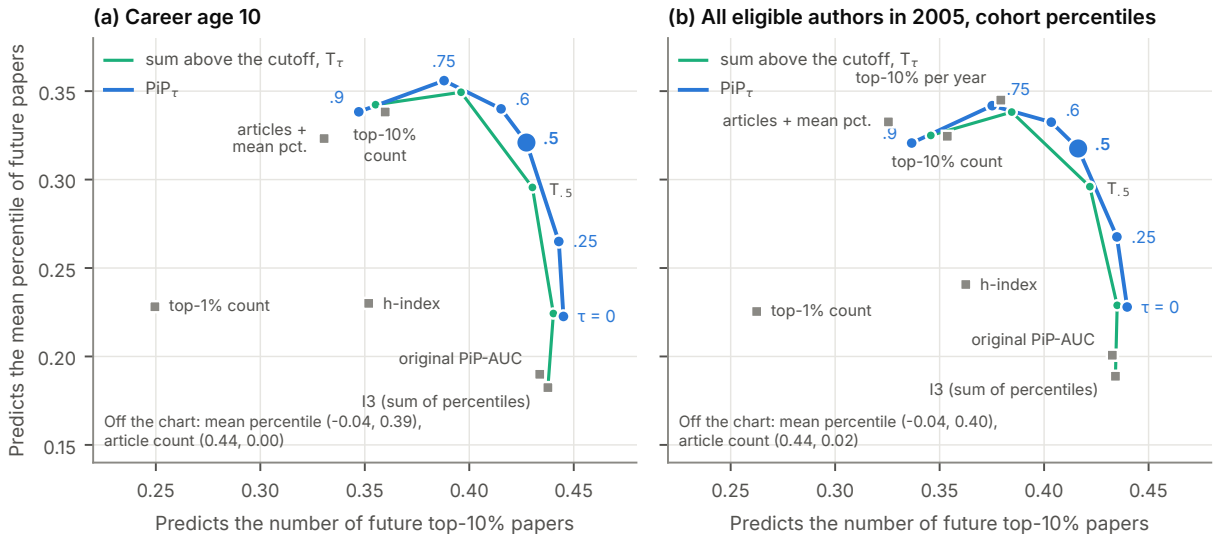


Figure 6: Indicators placed by how well they predict the number of a scholar's future top-10% papers (horizontal) and the mean percentile of their future papers (vertical), Spearman correlations. The blue curve is PiP at cutoffs from $\tau = 0$ to 0.9; the large dot is $\tau = 0.5$. The green curve is the sum of percentiles above the cutoff, T_τ, at $\tau = 0.25$, 0.5, 0.75 and 0.9, starting from I3 ($\tau = 0$). Grey squares are other indicators built from the same paper percentiles, the *h*-index and the original PiP-AUC; "articles + mean pct." averages the cohort percentiles of the article count and of the mean percentile. (a) Eligible authors at career age 10, raw values. (b) All eligible authors in 2005 at career ages 2–30, within-cohort percentiles (the top-10% count per year of career is raw). For the third outcome, a future top-1% paper, PiP at every cutoff from 0 to 0.75 has a higher AUC than every grey indicator in both settings.

article count and mean percentile, beats PiP on at most one of the three outcomes, and none beats it on the chance of a future top-1% paper. This describes a trade-off among these three outcomes, not a ranking for every purpose: which indicator is preferable depends on how an evaluator weighs them.

Figure 6 places the indicators by how well they predict future quantity and future citation impact, with PiP and T_τ at every cutoff. The PiP cutoffs trace a frontier. With $\tau = 0$, PiP predicts at least as well as I3 and the original PiP-AUC on all three outcomes in every setting (with $\tau = 0.25$, in all but one); with $\tau = 0.75$ it does at least as well as the top-10% count in every setting; and from $\tau = 0.25$ to 0.75 it does better than the *h*-index on all three. The T_τ curve runs just inside the PiP curve and meets it near $\tau = 0.9$; at cutoffs from 0.25 to 0.75 it is close to PiP on future quantity and lower on future citation impact. Three indicators lie on or beyond the frontier. The mean percentile lies beyond it in the direction of pure citation impact, where it gives up any information about quantity. The article count lies just beyond it in two settings (at age 15 and in 2000), in the direction of pure quantity. The top-10% count per year of career, which differs from the plain count only when seniorities are mixed, lies on the curve next to $\tau = 0.75$. For these two outcomes, the cutoff therefore selects a point on a curve that lies on or outside the other combinations of paper percentiles we tested, apart from the two extremes, and $\tau = 0.5$ sits near its knee: from $\tau = 0.25$ to 0.5 the curve gains 0.056 in future citation impact for 0.016 in quantity, and beyond 0.5 each gain in impact costs more in quantity.

Replication across cohorts. With more than a million authors per setting, sampling error is small: the standard error of a Spearman correlation of 0.4 would be below 0.001 for independent observations. Authors

Table 5: How much earlier PiP reaches each indicator's accuracy, in years. For each target age T from 5 to 10, we find the (interpolated) age at which PiP first matches the indicator's accuracy at T; the table gives the range of T minus that age. Authors eligible at career age 3, outcome at career ages 11–20.

Compared with	Precision for future stars	AUC for future stars	AUC for a future top-1% paper
Citations	1.7–3.2	1.4–2.8	1.2–2.6
h-index	1.2–1.8	1.1–1.6	1.3–1.9
Articles	0.7–1.6	0.3–0.9	0.6–1.3
Articles + mean pct.	1.9–4.5	1.7–4.0	1.7–4.1
Top-10% count	0.5–0.6	1.0–1.3	0.8–1.3
I3	0.4–0.7	0.2–0.4	0.4–0.8
$T_{0.5}$ (sum above the cutoff)	0.1–0.2	0.0	0.1–0.3
Original PiP-AUC	0.0–0.2	0.0–0.2	0.3–0.4

share papers, so the observations are not independent and this figure is optimistic. The more relevant question is whether the differences hold in different populations. We repeated the age-10 comparison separately for each start year from 1975 to 2000, 26 cohorts of 17,867 to 122,114 authors. PiP beats the h-index on all three outcomes in all 26 cohorts, by 0.06–0.10 on the number of future top papers. It beats the top-10% count on the number of future top papers (by 0.05–0.08) and on the top-1% outcome in all 26 cohorts, and falls behind it on future citation impact in 20 (by up to 0.07). Against I3, the article count and the original PiP-AUC, PiP wins on the top-1% outcome and on future citation impact in every cohort, and on the number of future top papers in 11 to 14 of the 26, with differences between -0.03 and $+0.04$. PiP is the best predictor of a future top-1% paper in every setting.

5.6 Early detection

The settings above compare indicators measured at one age. Early detection asks a different question: how soon does each indicator pick out the scholars who will do well later? To answer it we fix both the population and the outcome, and vary only the age at which the indicators are measured. The population is every author who is eligible at career age 3 (306,854 authors with start years 1975–2000); since eligibility persists, each of them has every indicator at every later age. The outcome is their work at career ages 11–20, scored as before (each article at its own age 5). We call an author a *future star* if they are in the top 10% of their start-year cohort, within the population, by the number of top-10% articles published at ages 11–20 (ties broken by the mean percentile of those articles). We then measure each indicator at career ages 3, 4, . . . , 10, and ask how well it finds the future stars: the precision of picking the top 10% of each cohort by the indicator (chance is 0.10), and the AUC for the same outcome. Authors who tie at the boundary of the top 10%, as they often do on the h-index, each count with the share of the remaining places, which gives the expected precision under random tie-breaking. We also report the AUC for publishing at least one top-1% article at ages 11–20. Each indicator enters as its percentile within the start-year cohort.

PiP is the most accurate indicator at every measurement age on both AUC measures, and on precision at every age but 9 and 10, where the original PiP-AUC matches it to within 0.001 (Figure 7). The sum of percentiles above the cutoff, $T_{0.5}$, is never more accurate than PiP but stays within 0.006 of it at every age. Measured at career age 3, the top 10% of a cohort by PiP contains 21% future stars, twice the chance rate,

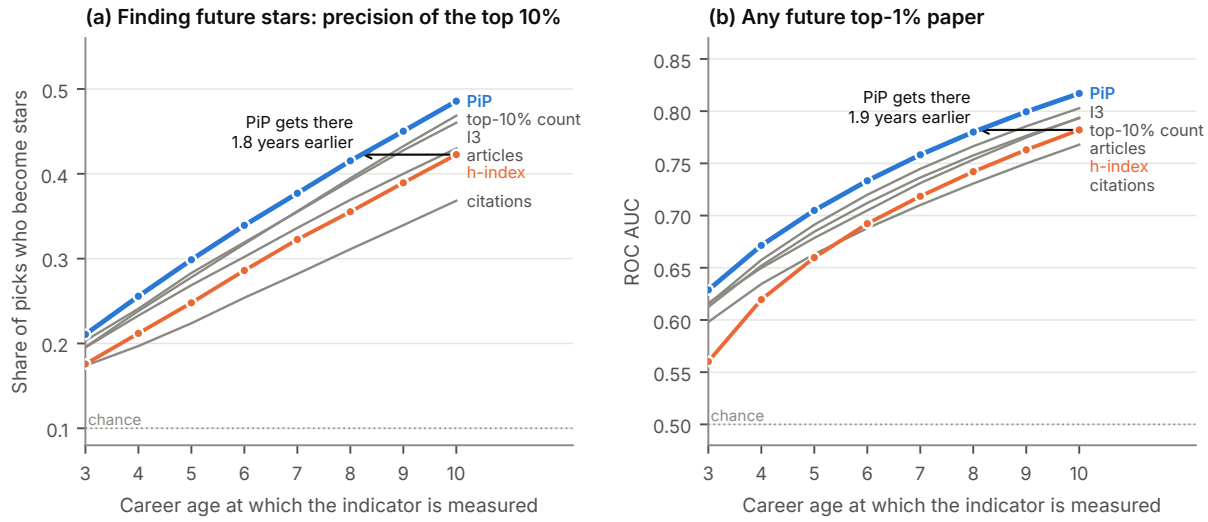


Figure 7: Early detection. Authors eligible at career age 3 (306,854; start years 1975–2000), with each indicator measured at career ages 3 to 10 and the outcome fixed: their articles published at career ages 11–20. (a) Of the authors in the top 10% of their cohort by the indicator (ties at the boundary shared at random), the share who become future stars (the top 10% of the cohort by the number of top-10% articles at ages 11–20). (b) ROC AUC for at least one top-1% article at ages 11–20. The arrow marks the career age at which PiP reaches the accuracy that the *h*-index has at age 10. The original PiP-AUC and the sum of percentiles above the cutoff, $T_{0.5}$, not shown, lie just below or level with PiP in both panels.

against 18% for the *h*-index; at age 5 the shares are 30% and 25%, and at age 10, 49% and 42%. The AUC for a future top-1% paper is 0.63 against 0.56 at age 3 and 0.71 against 0.66 at age 5. Because every indicator improves with age, the gaps translate into time (Table 5): PiP reaches the accuracy that the *h*-index has at age 10 1.6 to 1.9 years earlier, at about age 8, and the accuracy of the citation count 2.6 to 3.2 years earlier. Its lead over the article count, the top-10% count and I3 is smaller, at most 1.6 years, and the original PiP-AUC and $T_{0.5}$ are close behind (within 0.4 years). For the larger population of authors eligible at age 5 (711,986), PiP reaches the *h*-index's age-10 accuracy 1.5 to 1.8 years earlier and the citation count's 2.3 to 2.9 years earlier.

Without the h-index condition. Requiring an *h*-index of three by career age 3 keeps only 21% of the authors who have three articles by then, and it selects the population with one of the indicators being compared. We therefore repeated the analysis for all 1,455,891 authors with at least three articles at career age 3, with peers defined by the same rule. PiP still leads the *h*-index at every measurement age: the top 10% by PiP contains 24% future stars at age 3 against 20% for the *h*-index, and 31% against 27% at age 5, and PiP reaches the *h*-index's age-10 accuracy 1.2 to 1.7 years earlier. Its lead over the indicators closest to it is confined to early ages: I3, $T_{0.5}$ and the original PiP-AUC match or exceed it on the AUC for future stars from about career age 6, and $T_{0.5}$ and the original PiP-AUC match it on precision from about age 7, while PiP keeps the highest AUC for a future top-1% paper at every age.

On the other two outcomes of Section 5.5 the ordering at each age matches the one found there: the article count predicts the number of future top-10% papers slightly better than PiP at the earliest ages, and the mean

Table 6: ROC AUC for holding a senior honor. Columns 2–5: indicators measured at career age 5 or 10 in Semantic Scholar, honors received later; all eligible authors (the comparison uses every honoree and a 10% random sample of the others), only people with a Wikidata item, and only our Google Scholar profiles matched to Semantic Scholar (mostly faculty). Columns 6–7: current Google Scholar scores of 15,890 profiled scholars against holding an honor and having an English Wikipedia article (agreement, not prediction). PiP's bootstrap intervals are about ±0.015 for all authors and ±0.02 within the smaller groups; differences between indicators are estimated more precisely because they are paired.

	Predicting later honors (Semantic Scholar)				Google Scholar, now	
Future honorees / positives	All, age 5 1,064	All, age 10 1,555	Wikidata, 10 751	GS faculty, 10 890	Honors 1,497	Wikipedia 2,160
PiP ($\tau = 0.5$)	0.749	0.816	0.628	0.631	0.712	0.716
PiP ($\tau = 0.75$)	**0.760**	**0.826**	**0.646**	**0.638**	0.711	**0.721**
Original PiP-AUC	0.719	0.791	0.603	0.622	0.696	0.692
h-index	0.710	0.782	0.610	0.623	0.708	0.704
Citations	0.701	0.789	0.629	0.627	**0.714**	**0.721**
Articles	0.693	0.772	0.569	0.611	0.677	0.675
Top-10% count	0.747	0.816	0.644	0.634	0.709	0.720
I3	0.721	0.788	0.595	0.622	0.696	0.692
Mean percentile	0.527	0.568	0.570	0.535	0.611	0.611

percentile predicts their citation impact best. PiP's advantage in early detection is in finding the scholars at the top, who combine many papers with highly cited ones, and in anticipating exceptional papers. Most of this advantage comes from year-normalized percentiles with a cutoff, which $T_{0.5}$ shares; the comparison with peers at every threshold adds little to early detection.

5.7 Later recognition

The tests so far judge citation indicators by later citations. As a check against an outcome that is not a citation count, we ask whether the indicators anticipate later scientific honors. We take honors from Wikidata: 65 senior honors given for research, namely membership of national academies (such as the US National Academy of Sciences, the Royal Society and the Leopoldina), fellowships of learned societies (AAAS, ACM, IEEE, APS and others) and major prizes and fellowships (such as the Guggenheim, MacArthur and Gairdner awards). We leave out plain society memberships, state orders, student scholarships and the citation-based Clarivate Citation Laureates. Honors reach Semantic Scholar authors in two ways: through Wikidata items that carry a Semantic Scholar author ID (41,272 items), and through items that carry a Google Scholar ID (12,434 honorees), which we link to Semantic Scholar with the title matching of Section 5.9. We measure each indicator at career age 5 or 10, as a percentile within the start-year cohort (start years 1960–2000), and drop people whose first dated honor came at or before that age; most honors come decades later. This leaves 1,064 future honorees at age 5 and 1,555 at age 10.

Among all eligible authors, PiP measured at career age 5 picks out the future honorees with an AUC of 0.75, against 0.71 for the *h*-index and 0.70 for citations; at age 10, 0.82 against 0.78 and 0.79 (Table 6). The paired bootstrap intervals for PiP's lead over the *h*-index exclude zero at both ages (+0.039 [0.029, 0.049] and +0.035 [0.030, 0.040]). PiP ties the top-10% count (+0.000 [−0.005, 0.005] at age 10), and the impact end of the dial does better than the default: at $\tau = 0.75$ the AUC is 0.76 and 0.83. Recognition rewards highly

cited work more than volume: the article count, I3 and the original PiP-AUC, which lean towards volume, do worst apart from the mean percentile.

Within groups that are already notable, every indicator separates future honorees only weakly (AUC 0.59–0.65). There, PiP is ahead of the h-index at career age 10 (by 0.018 [0.010, 0.025] among people with a Wikidata item and 0.008 [0.003, 0.014] among the Google Scholar faculty) and level with it at age 5, while the top-10% count and PiP at $\tau = 0.75$ or higher do best. Restricting to honors with a recorded date gives the same pattern. On the Google Scholar profiles, where scores and honors are measured at the same time, the indicators are close: PiP is slightly ahead of the h-index for holding an honor ($+0.004$ [0.001, 0.008]) and for having a Wikipedia article ($+0.012$ [0.009, 0.015]), and total citations and the top-10% count do as well or better.

The recognition test therefore supports PiP over the h-index with an outcome outside citation counts, most clearly for detecting future honorees among all scholars and early in their careers. Within the populations closer to faculty evaluation the gain over the h-index is small, and the top-10% count does as well. The test also shows that the default cutoff is not the best choice for every purpose: for anticipating recognition, an evaluator should move the dial towards impact. Several caveats apply. Wikidata covers honors unevenly across countries and fields, so an author without a recorded honor is not a confirmed non-recipient; older cohorts have had more time to receive senior honors; and honors depend on nominations, networks and opportunity as well as on the work. These processes can strengthen or weaken the association between an indicator and recorded honors, so the AUCs measure agreement with recorded recognition, not with merit.

5.8 Fields

We assign each author the most frequent field among their articles by career age 10 (whole counting) and repeat the analyses by field. Fifteen fields have at least 3,000 eligible authors with start years 1975–2000. In every one of them, PiP predicts better than the h-index on all three outcomes; it predicts future top-1% papers and future citation impact better than the original PiP-AUC in every field, while the original form is better on the number of future top-10% papers in about half of them, as it is overall. For example, in economics PiP reaches 0.39 / 0.74 / 0.27 on the three outcomes against 0.31 / 0.67 / 0.17 for the h-index, and in mathematics 0.38 / 0.75 / 0.28 against 0.29 / 0.68 / 0.19.

Pooling fields in the paper percentiles favors fields with high citation rates. A field-neutral score would put 10% of each field's authors in the top 10% of their cohort; with pooled percentiles the share ranges from 1.4% in education and 2.1% in economics to 14.7% in physics (Figure 8). We therefore also compute a field-normalized PiP, in which a paper's percentile is computed within its publication year and the author's main field (when the paper carries that field; otherwise the pooled percentile is kept), and peers are restricted to authors with the same main field and start year. For 1.45 million authors at career age 10, this brings every field's share of the cohort top 10% to between 6.9% and 11.6%. Within a field the ranking largely holds: the correlation between the pooled and the field-normalized score is 0.91–0.99 per field. We have not separated the sources of the remaining spread. Papers outside the author's main field keep their pooled percentile, and fields differ in how often peers tie, which also shapes the distribution of the score. The rule also gives a coauthored paper different percentiles for coauthors with different main fields, a choice that a rule based on

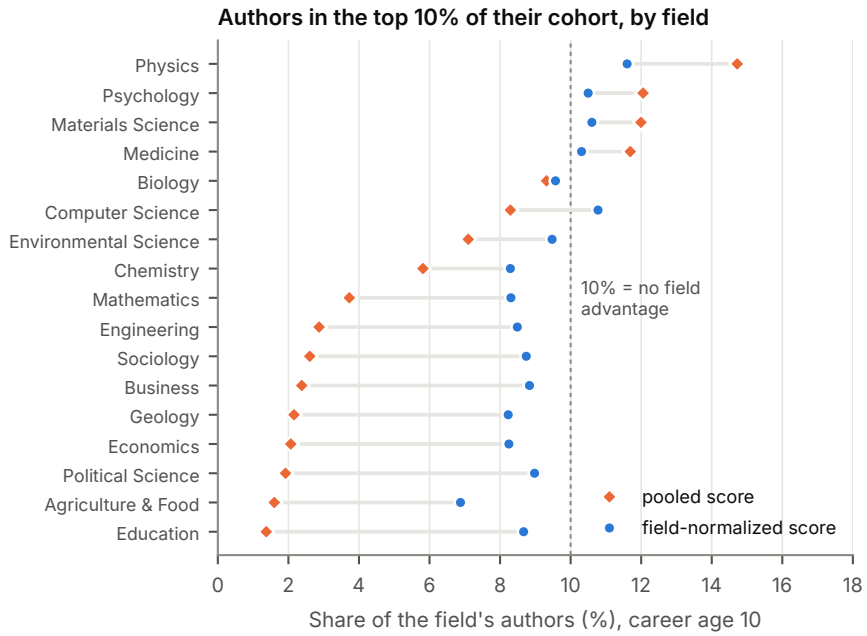


Figure 8: Share of each field's authors who are in the top 10% of their start-year cohort at career age 10, with PiP computed from pooled percentiles and with the field-normalized version. Fields with at least 5,000 authors.

the paper's own fields would avoid.

The two versions answer different questions. With outcomes measured on pooled percentiles, the pooled score predicts best (0.427 / 0.772 / 0.321 against 0.414 / 0.770 / 0.298); with field-normalized outcomes, the field-normalized score predicts best (0.407 / 0.772 / 0.316 against 0.405 / 0.769 / 0.300). Both remain far ahead of the *h*-index (0.333 / 0.717 / 0.209 on field-normalized outcomes). For comparisons across fields we recommend the field-normalized version; within a field the two are nearly interchangeable.

5.9 Robustness

Self-citations. A citation counts as a self-citation when the citing and the cited paper share at least one author. These are 8.1% of all citations; such rates depend strongly on the definition and the citation window (Aksnes, 2003). We removed them and reran the whole pipeline: percentiles, peer groups (the *h*-index condition for eligibility is then applied without self-citations), scores and outcomes (Table 7 in the appendix). The predictions of the number of future top-10% papers and of a future top-1% paper move by at most 0.003 for PiP. The *h*-index loses slightly more (0.352 to 0.340 at age 10). The prediction of future citation impact drops more, and unevenly: by 0.018 to 0.030 for PiP, 0.013 to 0.019 for the original PiP-AUC and 0.005 to 0.009 for the *h*-index. Part of PiP's lead on this outcome therefore runs through authors citing their own work, but the lead remains (0.298 against 0.221 at career age 10), and the ordering of the indicators does not change.

Eligibility without the h-index condition. The main rule requires an h-index of three, which conditions on citations and on one of the compared indicators, and which excludes far more authors in some fields than in others (Section 3). Under a rule that requires three articles and nothing else, the same-age settings grow to 2.81, 2.54 and 2.00 million authors at career ages 5, 10 and 15 (from 1.17, 1.45 and 1.24 million). PiP still beats the h-index on all three outcomes at every age, by 0.03 to 0.06, half to two thirds of the margin under the main rule; at career age 10 it reaches 0.420, 0.805 and 0.493 against 0.376, 0.770 and 0.443. It again has the highest AUC for a future top-1% paper, and no other indicator, including $T_{0.5}$ and the mix of article count and mean percentile, beats it on more than one outcome. Section 5.6 reports early detection under the same rule.

Author disambiguation. Semantic Scholar's author records come from automatic disambiguation, which can split one person across several records or merge different people. We matched the 16,218 Google Scholar profiles to Semantic Scholar through their publications: a profile's work matches a Semantic Scholar paper on its normalized title (at least 25 characters) and a publication year within one year, and the profile's counterpart is the Semantic Scholar author with the same surname who holds the most matched works. We find a confident counterpart (at least five matched works and at least 30% of the profile's titles) for 14,511 profiles (89%). For the median person, 76% of their Google Scholar works are found in Semantic Scholar, and 90% of those found sit under the main author record. Splitting is common, though: 64% of people have at least two of their works under a second record with the same surname and first initial, and 42% under a third. The main record is smaller than the profile (a median 67% of the profile's cited works, and 78% of its h-index), and its start year is later than the profile's for 47% of people, because early work is missing or sits under another record.

Despite these differences, the two sources rank people similarly. For the 9,796 people who have scores in both sources (Semantic Scholar start year 1995 or later, evaluated in 2025), the Spearman correlation between sources is 0.82 for the h-index, 0.78 for the original PiP-AUC and 0.78 for PiP; each source computes percentiles against its own population (all eligible Semantic Scholar authors, and Google Scholar profiles). The agreement is lower for people split across three or more records (0.73) than for people with one record (0.76), but not by much. We read this as evidence that the results above are not artifacts of one disambiguation algorithm, and as a warning that individual Semantic Scholar scores can be off when a person's work is split.

Survivorship. The population includes every eligible author, not only those who stayed in research. Whether an author is still publishing at career ages 11–15 is predicted by their age-5 indicators with an AUC of 0.68 for the article count, 0.65 for the original PiP-AUC, 0.62 for PiP, 0.58 for the h-index and 0.54 for citations. Continuing to publish is a matter of volume, so the indicators that weigh volume most predict it best. The prediction results of Section 5.5 count authors who stop publishing as having no future top papers, which favors indicators that predict survival; PiP still leads the article count and the original PiP-AUC, the two that predict survival best, on the top-1% outcome and on future citation impact.

6 Discussion

Why PiP improves on the h-index. The h-index summarizes a record by one point; PiP uses every paper above the cutoff, through a comparison with peers at every impact threshold. Three properties explain most of the gain in prediction. First, PiP gives credit to the best papers in proportion to how rare they are among peers: a single top paper raises the standing at the highest thresholds, where most peers have none, and the h-index ignores everything above the corner. Second, it gives credit to recent papers as soon as they stand out among papers of the same age, while the h-index waits until they have accumulated enough raw citations; this matters most early in a career, when most papers are recent, and is the likely source of much of PiP's lead in early detection (Section 5.6). The sum of percentiles above the cutoff, $T_{0.5}$, shares this property and detects future high performers almost as early, which fits this reading without isolating it. Third, its scale does not depend on the cohort, so scholars of different seniority can be compared, while the raw h-index favors those who started later, when citation rates were higher, and those who have been active longer. The comparison with T_τ also separates the parts of the construction. Year-normalized percentiles with a cutoff account for most of the gain over the h-index in predicting how many top papers will follow and in early detection; the comparison with peers at every threshold, the first property, adds to predicting the citation impact of future papers and the chance of an exceptional one.

Prediction and cumulative advantage. Citations reward visibility as well as research quality: well-known scholars attract more citations for comparable work (Merton, 1968; Petersen et al., 2014). Part of the predictive power of any citation-based indicator, PiP and the h-index alike, may therefore come from reputation feeding itself rather than from the persistence of research quality. Removing self-citations, the most mechanical channel, leaves the ranking of indicators unchanged and lowers PiP's prediction of future citation impact by 0.02 to 0.03 and the h-index's by less than 0.01; it does not rule out the broader effect of reputation, which would need an external source of variation to separate. Indicators can absorb cumulative advantage to different degrees, so it may also account for part of the differences between them. The test against later honors (Section 5.7) uses an outcome that is not a citation count and gives the same ordering of PiP and the h-index, although honors are themselves shaped by visibility.

Fields and windows. The pooled score favors fields with high citation rates, and the field-normalized version removes most of that advantage. Field normalization brings its own choices: the field scheme, the assignment of multi-field papers and the assignment of an author to a main field (Zitt et al., 2005; Waltman and van Eck, 2015, 2019). Semantic Scholar's 23 fields are broad, and subfields with different citation habits share a reference set, so field normalization removes only the differences between these broad fields. Percentiles of recent papers are also noisier in fields where citations accumulate slowly (Section 5.1). In those fields, an evaluation might count only papers older than two or three years, at the cost of ignoring the most recent work.

Limitations.

- *Disambiguation.* Split author records shorten careers and move start years, which places the affected person in a later cohort with a partial record. For evaluating individuals, a curated record such as a Google Scholar or ORCID profile should be used.

- *Whole counting.* Every co-author receives full credit for every paper, regardless of the number of authors or their roles. A fractional version would weight each paper's contribution to $n_a(\theta)$ by the author's share (Waltman and van Eck, 2015); we have not tested it.

- *The peer group.* Peers are defined by the year of the first article. Authors who start late, interrupt their careers or move between research and other work are compared with a cohort whose careers ran differently. The eligibility rule (three articles and an h-index of three) excludes much of the earliest stage of a career and of the fields that cite slowly; without the h-index condition the main comparisons hold with smaller margins (Sections 5.6 and 5.9). A scholar with one exceptional paper and nothing else, the example of Section 4.4, is outside both rules; the score can be computed for such a record against a cohort, but our tests do not cover it.

- *Retrospective reconstruction.* Publications and citations are limited to each evaluation year, but author records, publication types and fields come from the 2026 release; an evaluation at the time would have seen different records.

- *Reference set.* The article reference set depends on Semantic Scholar's publication types, which are incomplete; untyped records are excluded from both the reference sets and the counts.

- *Consistency.* PiP is not consistent in the sense of Waltman and van Eck (2012), although reversals are rare in simulation.

Responsible use. PiP is meant to inform comparisons among scholars at a similar career stage, alongside reading the work itself and expert judgment, as the Leiden Manifesto (Hicks et al., 2015) and DORA (San Francisco Declaration on Research Assessment, 2012) recommend. Its cutoff should be chosen before an evaluation, not tuned to favor a candidate, and the field version should be used when people from different fields are compared. A predictive correlation of 0.4 leaves most of a scholar's future unexplained.

7 Conclusion

PiP rates a scholar by the citation percentiles of their papers above a cutoff, compared with peers who started in the same year at every impact threshold. On 11.1 million Semantic Scholar authors its distribution changes little across cohorts since 1960, it is more stable over a career than the h-index, and it predicts future top papers and the citation impact of future work better than the h-index in every setting and field we tested; the lead narrows but remains when eligibility does not depend on citations. Among indicators built from the same paper percentiles it offers a balance rather than dominance: none of them beats it on more than one of three outcomes, and its comparison with peers at every threshold adds most to predicting the citation impact of future work. It reaches the h-index's accuracy in detecting future high performers 1.6 to 1.9 years earlier.

A single parameter moves it between quantity and impact. The construction needs only publication years and citation counts by year, and the code in the replication package computes it at full scale in a database. The website `https://www.pip-score.org` computes the original PiP-AUC from Google Scholar profiles.

Data and code availability

The SQL that builds every table in this paper, the numerical checks of the properties in Section 4, the data behind every figure and the plotting scripts are in the `paper/` directory of `https://github.com/ipeirotis-org/scholar_v2`. Semantic Scholar data are available under the ODC-BY license from the Semantic Scholar Datasets API.

Competing interests

Panos Ipeirotis developed and maintains `https://www.pip-score.org`, a free, non-commercial website that computes the original PiP-AUC from Google Scholar profiles. The authors have no other competing interests.

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A Self-citations

Table 7: Prediction results with and without self-citations (with / without). Self-citations are removed throughout: in the paper percentiles, the peer groups, the scores and the outcomes. Raw scores, pooled over start years; for the year 2005, Table 4 reports cohort percentiles, which differ slightly.

Setting	Indicator	Top-10% count	Top-1% AUC	Mean pct.
Age 5	PiP	.378 / .378	.715 / .712	.264 / .234
	Original PiP-AUC	.354 / .355	.683 / .681	.108 / .093
	h-index	.268 / .257	.646 / .640	.155 / .148
Age 10	PiP	.427 / .427	.772 / .771	.321 / .298
	Original PiP-AUC	.434 / .435	.756 / .755	.190 / .171
	h-index	.352 / .340	.719 / .714	.230 / .221
Age 15	PiP	.436 / .437	.799 / .799	.343 / .325
	Original PiP-AUC	.453 / .454	.792 / .790	.236 / .219
	h-index	.384 / .374	.758 / .754	.270 / .261
Year 2005	PiP	.407 / .407	.776 / .777	.310 / .291
	Original PiP-AUC	.434 / .437	.770 / .771	.201 / .188
	h-index	.284 / .277	.691 / .689	.174 / .169

B Prediction results in all settings

Table 8: Prediction of the next ten years of work in all five settings: Spearman correlation with the number of future top-10% articles / ROC AUC for at least one future top-1% article / Spearman correlation with the mean percentile of future articles. Same-year settings include career ages 2–30. PiP and the original PiP-AUC are raw scores pooled over start years; the h-index is given raw and as a cohort percentile.

Setting	PiP ($\tau = 0.5$, raw)	Original PiP-AUC (raw)	h-index (raw / cohort pct.)	
Age 5	.378 / .715 / .264	.354 / .683 / .108	.268 / .646 / .155	.256 / .647 / .178
Age 10	.427 / .772 / .321	.434 / .756 / .190	.352 / .719 / .230	.352 / .721 / .238
Age 15	.436 / .799 / .343	.453 / .792 / .236	.384 / .758 / .270	.385 / .760 / .272
Year 2000	.381 / .761 / .288	.398 / .751 / .186	.269 / .682 / .152	.328 / .721 / .232
Year 2005	.407 / .776 / .310	.434 / .770 / .201	.284 / .691 / .174	.363 / .737 / .241